📘 Chapter 29-VLAN Trunking,Native VLAN & Router on a Stick (ROAS)

🧠 1️⃣ Access Port vs Trunk Port

When configuring VLAN networks, **switch ports must be set to the correct mode**.

🖥️ Access Port

An **Access Port** carries traffic for **only ONE VLAN**.

✔ Used for **end devices**
✔ Example devices:

- PCs 💻
- Printers 🖨️
- IP Phones ☎️
- Cameras 📷

Example configuration:

switchport mode access
switchport access vlan 10

🌐 Trunk Port

A **Trunk Port** carries traffic for **MULTIPLE VLANs** over a single interface.

✔ Used between:

- Switch ↔ Switch
- Switch ↔ Router
- Switch ↔ Layer 3 Switch

Example configuration:

switchport mode trunk

🎯 Important Rule

Device Type	Port Mode
End Device (PC)	Access Mode
Network Device (Switch/Router)	Trunk Mode

📌 **Very important CCNA concept**

🧩 Example VLAN Topology

Imagine a network with **two switches**.

PCs → SW1 → SW2 → Router

Engineering department exists in **two locations**:

- Some PCs connected to **SW1**
- Some PCs connected to **SW2**

All belong to:

VLAN 10

Therefore the **switch-to-switch link must carry VLAN 10 traffic**.

This is why **TRUNK ports are required**.

🧠 Why Trunk Ports Are Needed

In small networks we could connect switches like this:

VLAN10 → Interface 1
VLAN20 → Interface 2
VLAN30 → Interface 3

But in real networks:

❌ Too many VLANs
❌ Wastes switch interfaces
❌ Routers may not have enough ports

Instead we use **TRUNK PORTS**.

🌉 Trunk Port Advantage

One interface can carry **multiple VLAN traffic simultaneously**.

Example:

Gig0/1 carries VLAN 10, 20, 30

🏷️ 2️⃣ VLAN Tagging

Question:

❓ If many VLANs share one trunk link, how does the switch know which VLAN a frame belongs to?

Answer:

VLAN TAG

Switch **adds a tag to the Ethernet frame**.

This tag contains the **VLAN ID**.

📦 Tagged vs Untagged Frames

Port Type	Frame Type
Trunk Port	Tagged
Access Port	Untagged

📌 Important concept for CCNA exam.

🧠 3️⃣ VLAN Trunking Protocols

There are **two trunk protocols**.

Protocol	Description
ISL	Cisco proprietary
IEEE 802.1Q	Industry standard

🔴 ISL (Inter-Switch Link)

- Cisco proprietary
- Older protocol
- Rarely used today

🟢 IEEE 802.1Q (Dot1Q)

- Industry standard
- Supported by almost all vendors
- Used in real networks

📌 **For CCNA you mainly study 802.1Q**

📦 4️⃣ Ethernet Frame with 802.1Q Tag

802.1Q inserts a **4-byte VLAN tag** into the Ethernet frame.

The tag is inserted between:

Source MAC
and
Type/Length

Tag size:

4 bytes (32 bits)

🧩 Components of 802.1Q Tag

The tag has two main sections.

1️⃣ TPID – Tag Protocol Identifier

- Size: **16 bits**
- Value: **0x8100**

This tells the switch:

This frame uses 802.1Q tagging

2️⃣ TCI – Tag Control Information

TCI contains **three fields**.

🔹 PCP (Priority Code Point)

Size:

3 bits

Used for:

Class of Service (CoS)

Allows prioritizing important traffic.

Example:

- Voice calls 📞
- Video streaming 🎥

🔹 DEI (Drop Eligible Indicator)

Size:

1 bit

Used when network is congested.

Frames marked with DEI may be dropped.

🔹 **VID (VLAN ID)**

Size:

12 bits

This field identifies the VLAN.

📊 VLAN ID Range

12 bits allow:

2^12 = 4096 VLANs

Range:

0 – 4095

However:

VLAN ID	Status
0	Reserved
4095	Reserved

Usable VLANs:

1 – 4094

📌 Important CCNA fact.

🔄 5️⃣ Inter-VLAN Routing

Devices in different VLANs **cannot communicate directly**.

Example:

Sales VLAN (30)
HR VLAN (20)

Switch **cannot route traffic between them**.

Traffic must go through:

Router

This process is called:

INTER-VLAN ROUTING

📡 Packet Flow Example

Example:

Sales PC → HR PC

Steps:

1️⃣ Packet goes to switch
2️⃣ Switch sees destination VLAN is different
3️⃣ Switch forwards packet to router
4️⃣ Router checks routing table
5️⃣ Router sends packet back to switch
6️⃣ Switch forwards to HR PC

📌 Important rule:

Switch never routes between VLANs

Only **router or Layer-3 switch** can do routing.

🧠 6️⃣ Native VLAN

802.1Q includes a feature called:

Native VLAN

Default native VLAN:

VLAN 1

Important behavior:

Frames in the **native VLAN are NOT tagged**.

📦 Native VLAN Example

If a switch receives an **untagged frame on a trunk port**:

Switch assumes:

Frame belongs to native VLAN

⚠️ Native VLAN Mismatch

If switches use **different native VLANs**, problems occur.

Example warning:

CDP-4-NATIVE_VLAN_MISMATCH

🔒 Security Best Practice

Network engineers recommend:

Change native VLAN to an unused VLAN

Example:

VLAN 1001

Reason:

Prevents VLAN hopping attacks.

🔄 7️⃣ Router on a Stick (ROAS)

Router on a Stick allows **multiple VLANs to use one router interface**.

Instead of multiple physical interfaces, we use:

SUBINTERFACES

Each subinterface represents one VLAN.

Example:

g0/0.10
g0/0.20
g0/0.30

Each has:

- VLAN tag
- IP address

📡 How ROAS Works

Router receives tagged frames.

Router checks VLAN tag.

Router forwards traffic through the correct subinterface.

🧪 VLAN Trunk + ROAS LAB

🧰 Devices Used

- 2 Switches
- 1 Router
- 7 PCs

🗺️ VLAN Design

VLAN	Department	Network
10	Engineering	10.0.0.0/26
20	HR	10.0.0.64/26
30	Sales	10.0.0.128/26

🧪 Step 1 – Create VLANs on Switches

SW1 Configuration

SW1> enable
SW1# configure terminal

vlan 10
name Engineering

vlan 20
name HR

vlan 30
name Sales

vlan 1001
name Native

🧪 Assign PC Ports (Access Mode)

```
interface fa0/1
switchport mode access
switchport access vlan 10

interface fa0/2
switchport mode access
switchport access vlan 10

interface fa0/3
switchport mode access
switchport access vlan 30

interface fa0/4
switchport mode access
switchport access vlan 30
```

Verify:

```
show vlan brief
```

SW2 Configuration

```
interface fa0/1
switchport mode access
switchport access vlan 20

interface fa0/2
switchport mode access
switchport access vlan 10

interface fa0/3
switchport mode access
switchport access vlan 10
```

🧪 Step 2 – Configure Trunk Between Switches

SW1

interface g0/1
switchport mode trunk
switchport trunk allowed vlan 10,20,30
switchport trunk native vlan 1001

Verify:

show interface trunk

SW2

interface g0/1
switchport mode trunk
switchport trunk allowed vlan 10,20,30
switchport trunk native vlan 1001

Trunk to Router

interface g0/2
switchport mode trunk
switchport trunk allowed vlan 10,20,30
switchport trunk native vlan 1001

🧪 Step 3 – Router on a Stick Configuration

Enable Router Interface

R1> enable
R1# configure terminal

interface g0/0
no shutdown

Subinterface for VLAN 10

interface g0/0.10
encapsulation dot1q 10
ip address 10.0.0.62 255.255.255.192

Subinterface for VLAN 20

interface g0/0.20
encapsulation dot1q 20
ip address 10.0.0.126 255.255.255.192

Subinterface for VLAN 30

interface g0/0.30
encapsulation dot1q 30
ip address 10.0.0.190 255.255.255.192

Verify Router Interfaces

show ip interface brief

🧪 Step 4 – Connectivity Test

Test ping between VLANs.

Example:

PC1 ping 10.0.0.65

If successful:

✓ Inter-VLAN routing works.

🔍 Packet Tracer Testing

Use:

Simulation Mode

Steps:

1️⃣ Enable ICMP filter
2️⃣ Send ping
3️⃣ Use **Capture/Forward**

You will see packets travel:

PC → Switch → Router → Switch → PC

🧠 Step-by-Step Lab Summary

Step 1

Create VLANs on switches.

Assign PC ports as **Access ports**.

Step 2

Configure **Trunk ports** between switches.

Allow only required VLANs.

Set **unused VLAN as native VLAN**.

Step 3

Configure **Router on a Stick**.

Create subinterfaces.

Assign gateway IP addresses.

Step 4

Test connectivity using **Ping**.

🌍 Real-World Example

Large company network:

Departments:

- Engineering
- HR
- Sales

Each department uses **separate VLANs**.

But only **one trunk link** connects switches.

Router performs **inter-VLAN routing**.

This reduces:

✓ Cable usage
✓ Hardware cost
✓ Network complexity

🎯 Interview One-Liners

Trunk Port

"A trunk port carries traffic for multiple VLANs using VLAN tagging."

802.1Q

"802.1Q is an IEEE standard used for VLAN tagging on trunk links."

Router on a Stick

"Router on a Stick uses router subinterfaces to route traffic between multiple VLANs using a single physical interface."

🧠 **Final Key Points**

✓ Access ports carry **one VLAN**
✓ Trunk ports carry **multiple VLANs**
✓ 802.1Q adds **VLAN tags**
✓ VLAN ID range = **1 – 4094**
✓ Native VLAN default = **1**
✓ Untagged frames belong to **native VLAN**
✓ Inter-VLAN communication requires **router or Layer 3 switch**
✓ Router on a Stick uses **subinterfaces**

🧠 1️⃣ Chapter Summary

🔹 Access Port

An **Access Port** is a switch port that carries **traffic for only one VLAN**.

📌 Used for **end devices** such as:

- 💻 PCs
- 🖨 Printers
- ☎ IP Phones
- 📷 Cameras

Example configuration:

switchport mode access
switchport access vlan 10

Important points:

✓ One VLAN only
✓ Frames are **untagged**
✓ Used for **end devices**

🌐 Trunk Port

A **Trunk Port** is a switch port that carries **traffic for multiple VLANs simultaneously**.

📌 Used between:

- Switch ↔ Switch
- Switch ↔ Router
- Switch ↔ Layer-3 Switch

Example configuration:

switchport mode trunk

Example:

Interface	VLANs
Gig0/1	10, 20, 30

This means the interface carries **traffic for multiple VLANs**.

🏷️ VLAN Tagging

When multiple VLANs share one trunk link, the switch must identify which VLAN the frame belongs to.

This is done using:

VLAN Tagging

The switch inserts a **VLAN tag inside the Ethernet frame**.

This tag includes the **VLAN ID**.

🧩 Tagged vs Untagged Frames

Port Type	Frame Type
Access Port	Untagged
Trunk Port	Tagged

📌 Exception: **Native VLAN frames are untagged even on trunk ports**

🌉 VLAN Trunking Protocols

There are two trunking protocols.

Protocol	Description
ISL	Cisco proprietary
802.1Q	Industry standard

🔴 ISL (Inter-Switch Link)

- Cisco proprietary protocol
- Old technology
- Rarely used today

🟢 IEEE 802.1Q (Dot1Q)

- Industry standard
- Used in almost all modern networks
- Inserts **4-byte VLAN tag**

📦 802.1Q Tag Structure

802.1Q adds **4 bytes to Ethernet frame**.

Main components:

Field	Size	Purpose
TPID	16 bits	Tag protocol identifier
PCP	3 bits	Priority
DEI	1 bit	Drop eligibility
VID	12 bits	VLAN ID

📊 VLAN ID Range

VLAN ID	Status
0	Reserved
1	Default VLAN
4095	Reserved

Usable VLANs:

1 – 4094

🔄 Inter-VLAN Routing

Devices in different VLANs **cannot communicate directly**.

Example:

- VLAN 10 (Engineering)
- VLAN 20 (HR)

A **Layer-2 switch cannot route between them**.

Traffic must go through:

✓ Router
✓ Layer-3 switch

This process is called:

Inter-VLAN Routing

🧠 Native VLAN

The **Native VLAN** is a special VLAN on trunk links.

Important behavior:

✓ Frames from native VLAN are **NOT tagged**

Default native VLAN:

VLAN 1

⚠ Native VLAN Mismatch

If two switches use **different native VLANs**, this error occurs:

CDP-4-NATIVE_VLAN_MISMATCH

This can cause:

❌ VLAN leakage
❌ Security problems

🔒 Security Best Practice

Change native VLAN to **unused VLAN**.

Example:

VLAN 1001

Purpose:

✓ Prevent VLAN hopping attacks

🔄 Router on a Stick (ROAS)

Router on a Stick allows **multiple VLANs to use one router interface**.

Instead of using multiple physical ports, the router creates **subinterfaces**.

Example:

Subinterface	VLAN
g0/0.10	VLAN 10
g0/0.20	VLAN 20
g0/0.30	VLAN 30

Each subinterface has:

✓ VLAN tag
✓ Gateway IP address

📡 How ROAS Works

1️⃣ PC sends packet
2️⃣ Switch forwards packet to router trunk
3️⃣ Router receives tagged frame

4️⃣ Router routes packet between VLANs
5️⃣ Packet returned to switch
6️⃣ Switch forwards to destination

🎯 Chapter Conclusion

This chapter explains **how VLAN traffic travels across switches and routers**.

Key ideas:

✓ Access ports connect end devices
✓ Trunk ports carry multiple VLANs
✓ 802.1Q tagging identifies VLANs
✓ Native VLAN frames are untagged
✓ Different VLANs require routing
✓ Router on a Stick enables inter-VLAN routing using one interface

📌 In real enterprise networks, **VLAN trunking and ROAS are fundamental concepts for scalable network design.**

🎤 Q & A

1️⃣ What is a Trunk Port?

Answer

A **Trunk Port** is a switch port that carries **traffic for multiple VLANs simultaneously**.

It uses **VLAN tagging (802.1Q)** to identify which VLAN the frame belongs to.

Example use cases:

- Switch to Switch connection
- Switch to Router connection

Example configuration:

interface g0/1
switchport mode trunk

2️⃣ What is an Access Port?

Answer

An **Access Port** is a switch port that belongs to **only one VLAN**.

It connects **end devices**.

Examples:

- PC
- Printer
- IP phone

Configuration example:

switchport mode access
switchport access vlan 10

3️⃣ What is VLAN Tagging?

Answer

VLAN tagging is a process where the switch inserts a **VLAN identifier into an Ethernet frame**.

This tag allows trunk links to carry **multiple VLAN traffic simultaneously**.

The tagging protocol used is:

IEEE 802.1Q

4️⃣ What is IEEE 802.1Q?

Answer

IEEE 802.1Q is the industry standard protocol used for **VLAN tagging on trunk links**.

It inserts a **4-byte VLAN tag inside Ethernet frames**.

Purpose:

✓ Identify VLAN ID
✓ Support multiple VLANs on one link

5️⃣ What is ISL?

Answer

ISL (Inter-Switch Link) is a Cisco proprietary VLAN tagging protocol.

Characteristics:

- Developed by Cisco
- Encapsulates Ethernet frames
- Rarely used today

Modern networks use **802.1Q instead**.

6️⃣ What is the Native VLAN?

Answer

The **Native VLAN** is a special VLAN on trunk ports where frames are sent **without VLAN tags**.

Default native VLAN:

VLAN 1

Important rule:

Untagged frames on trunk ports belong to **Native VLAN**.

7️⃣ What is Native VLAN Mismatch?

Answer

Native VLAN mismatch occurs when two trunk ports use **different native VLANs**.

Example:

Switch	Native VLAN
SW1	1
SW2	100

This causes:

❌ Network errors
❌ VLAN leakage
❌ Security risks

Error message example:

CDP-4-NATIVE_VLAN_MISMATCH

8️⃣ What is Router on a Stick?

Answer

Router on a Stick (ROAS) is a method used for **inter-VLAN routing** using a **single router interface**.

Instead of using multiple physical interfaces, the router creates **subinterfaces**.

Example:

interface g0/0.10
encapsulation dot1q 10
ip address 10.0.0.62 255.255.255.192

Each subinterface represents one VLAN.

9️⃣ Why Router on a Stick is used?

Advantages:

✓ Saves router interfaces
✓ Reduces hardware cost
✓ Simplifies network design
✓ Supports multiple VLANs

🔟 What is Inter-VLAN Routing?

Answer

Inter-VLAN routing is the process of **allowing communication between different VLANs**.

Example:

Engineering VLAN → HR VLAN

Devices in different VLANs require:

✓ Router
✓ Layer-3 Switch

1️⃣1️⃣ What command verifies trunk configuration?

show interface trunk

Displays:

- Trunk interfaces
- Allowed VLANs
- Native VLAN

1️⃣2️⃣ What command shows router subinterfaces?

show ip interface brief

This shows:

✔ Interface status
✔ IP addresses
✔ Subinterfaces

1️⃣3️⃣ Troubleshooting Question

Question

PC in VLAN 10 cannot ping PC in VLAN 20. Why?

Possible reasons:

1️⃣ Router not configured
2️⃣ Trunk not configured
3️⃣ VLAN not allowed on trunk
4️⃣ Incorrect gateway on PCs
5️⃣ Router subinterface missing

1️⃣4️⃣ Interview Scenario Question

Question

Why can't a switch route traffic between VLANs?

Answer

A **Layer-2 switch operates at OSI Layer 2** and forwards frames using **MAC addresses only**.

Routing requires:

- IP processing
- Layer 3 functionality

Therefore **router or Layer-3 switch is required**.

⭐ Important Tip

Explain Router on a Stick in one line

Answer:

"Router on a Stick is a method of inter-VLAN routing where a single router interface uses multiple subinterfaces with 802.1Q tagging to route traffic between VLANs."

🎯 Final Key Points to Remember

✓ Access ports carry **single VLAN**
✓ Trunk ports carry **multiple VLANs**
✓ 802.1Q adds **4-byte VLAN tag**
✓ VLAN ID range = **1–4094**
✓ Native VLAN frames are **untagged**
✓ Native VLAN default = **VLAN 1**
✓ Inter-VLAN routing requires **router or L3 switch**
✓ Router on a Stick uses **subinterfaces**
